# CFD_HW_5

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## Question 14: Exact Solution of Finite-Difference Scheme

### （a）

The original Lagrangian interpolation scheme is given by:

$$u_j^{n+1} = -\frac{(\Delta x - a\Delta t)a\Delta t}{2(\Delta x)^2}u_{j-2}^n + \frac{(2\Delta x - a\Delta t)a\Delta t}{(\Delta x)^2}u_{j-1}^n + \frac{(\Delta x - a\Delta t)(2\Delta x - a\Delta t)}{2(\Delta x)^2}u_j^n \tag{1}$$

Let the CFL to be $C \equiv \frac{a\Delta t}{\Delta x}$:

$$u_j^{n+1} = -\frac{C(1-C)}{2}u_{j-2}^n + C(2-C)u_{j-1}^n + \frac{(1-C)(2-C)}{2}u_j^n \tag{2}$$

Rearranging in the order of $C$:

$$u_j^{n+1} - u_j^n = -\frac{C}{2}[3u_j^n - 4u_{j-1}^n + u_{j-2}^n] + \frac{C^2}{2}[u_{j-2}^n - 2u_{j-1}^n + u_j^n] \tag{3}$$

Divide both sides by $\Delta t$ and substitute $C = \frac{a\Delta t}{\Delta x}$ back into the equation:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = -\frac{a}{2\Delta x}[3u_j^n - 4u_{j-1}^n + u_{j-2}^n] + \frac{a^2\Delta t}{2(\Delta x)^2}[u_{j-2}^n - 2u_{j-1}^n + u_j^n] \tag{4}$$

Rearranging the spatial derivative term to the left side yields the target equation:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + \frac{a}{2\Delta x}[3u_j^n - 4u_{j-1}^n + u_{j-2}^n] = \frac{a^2\Delta t}{2}\frac{[u_{j-2}^n - 2u_{j-1}^n + u_j^n]}{(\Delta x)^2} \tag{5}$$

### (b)

To perform the von Neumann stability analysis, we assume a discrete Fourier mode for the numerical solution:

$$u_j^n = G^n e^{ij\theta} \tag{6}$$

where $G$ is the amplification factor, $\theta = k\Delta x$ is the phase angle, and $i = \sqrt{-1}$.

Substituting this into $(2)$:

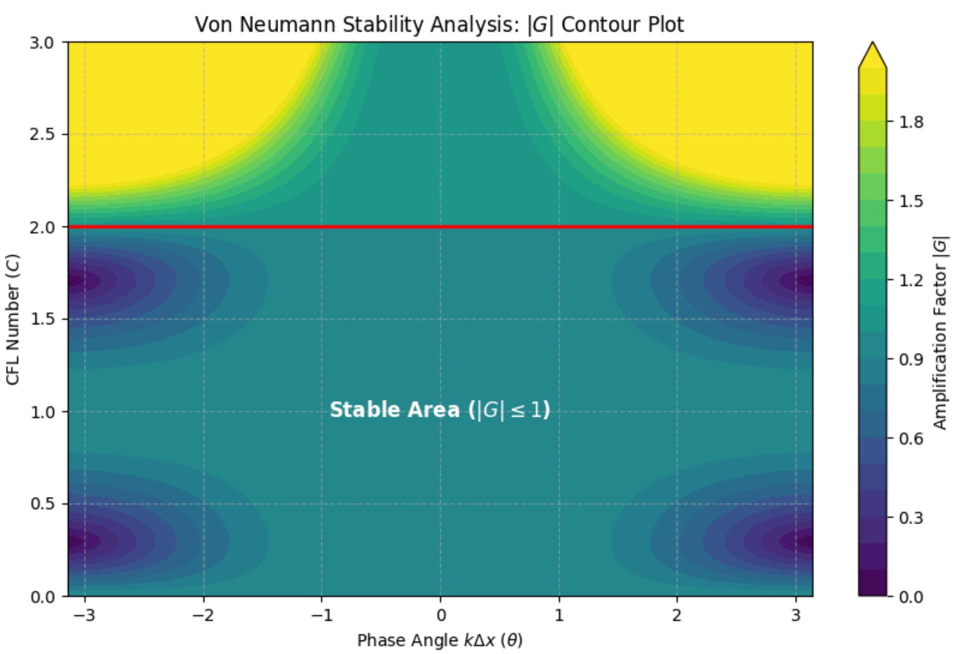
$$G = \frac{C(C-1)}{2} e^{-i2\theta} + C(2-C)e^{-i\theta} + \frac{(C-1)(C-2)}{2} \tag{7}$$

Using Euler's formula $e^{-i\theta} = \cos\theta - i\sin\theta$, we can separate $G$ into its real part ($G_R$) and imaginary part ($G_I$):

$$G_R = \frac{C(C-1)}{2}\cos(2\theta) + C(2-C)\cos\theta + \frac{(C-1)(C-2)}{2} \tag{8.1}$$

$$G_I = -\frac{C(C-1)}{2}\sin(2\theta) - C(2-C)\sin\theta \tag{8.2}$$

The magnitude of the amplification factor is $|G| = \sqrt{G_R^2 + G_I^2}$.




### (c)

The 1D advection equation has an exact solution. It means the wave simply moves at a constant speed $a$:

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = 0 \tag{9}$$

$$u(x, t + \Delta t) = u(x - a\Delta t, t) \tag{10}$$

On our grid, this means:

$$u_j^{n+1} = u(x_j - a\Delta t, t^n) \tag{11}$$

The error in our scheme comes from the interpolation step. If the point $x_j - a\Delta t$ lands exactly on a grid point that we already use (like point $j-1$ or $j-2$), we don't need to do any interpolation. In this case, the scheme has zero truncation error.

There are two special cases for $C = \frac{a\Delta t}{\Delta x}$:

**1. The point lands exactly on $j-1$:**

$$a\Delta t = \Delta x \implies C = 1 \tag{12}$$

If we plug $C = 1$ into our scheme, we get:

$$u_j^{n+1} = 0 \cdot u_{j-2}^n + 1 \cdot u_{j-1}^n + 0 \cdot u_j^n = u_{j-1}^n \tag{13}$$

This matches the exact solution.

**2. The point lands exactly on $j-2$:**

$$a\Delta t = 2\Delta x \implies C = 2 \tag{14}$$

If we plug $C = 2$ into our scheme, we get:

$$u_j^{n+1} = 1 \cdot u_{j-2}^n + 0 \cdot u_{j-1}^n + 0 \cdot u_j^n = u_{j-2}^n \tag{15}$$

This also matches the exact solution perfectly.

So, the two special values of $C$ is **1** and **2**.

# Question 15: Derivation of the Modified Equation

From Problem 14, our scheme looks like this:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + \frac{a}{2\Delta x}[3u_j^n - 4u_{j-1}^n + u_{j-2}^n] = \frac{a^2\Delta t}{2(\Delta x)^2}[u_{j-2}^n - 2u_{j-1}^n + u_j^n] \tag{1}$$

First, let's do Taylor expansions around the point $(x_j, t^n)$. To make it easier to write, I'll use subscripts for derivatives (like $u_x$ instead of $\frac{\partial u}{\partial x}$):

$$u_j^{n+1} = u + u_t\Delta t + \frac{1}{2}u_{tt}\Delta t^2 + \frac{1}{6}u_{ttt}\Delta t^3 + \frac{1}{24}u_{tttt}\Delta t^4 + \dots \tag{2}$$

$$u_{j-1}^n = u - u_x\Delta x + \frac{1}{2}u_{xx}\Delta x^2 - \frac{1}{6}u_{xxx}\Delta x^3 + \frac{1}{24}u_{xxxx}\Delta x^4 - \dots \tag{3}$$

$$u_{j-2}^n = u - 2u_x\Delta x + 2u_{xx}\Delta x^2 - \frac{4}{3}u_{xxx}\Delta x^3 + \frac{2}{3}u_{xxxx}\Delta x^4 - \dots \tag{4}$$

Plugging these back into the three parts of our scheme:

**1. Time derivative part:**

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = u_t + \frac{1}{2}u_{tt}\Delta t + \frac{1}{6}u_{ttt}\Delta t^2 + \frac{1}{24}u_{tttt}\Delta t^3 + \dots \tag{5}$$

**2. Spatial convection part:**

$$\frac{a}{2\Delta x}[3u_j^n - 4u_{j-1}^n + u_{j-2}^n] = au_x - \frac{a}{3}u_{xxx}\Delta x^2 + \frac{a}{4}u_{xxxx}\Delta x^3 + \dots \tag{6}$$

**3. Artificial diffusion part (right side):**

$$\frac{a^2\Delta t}{2(\Delta x)^2}[u_{j-2}^n - 2u_{j-1}^n + u_j^n] = \frac{a^2\Delta t}{2}u_{xx} - \frac{a^2\Delta t\Delta x}{2}u_{xxx} + \frac{7a^2\Delta t(\Delta x)^2}{24}u_{xxxx} + \dots \tag{7}$$

Putting it all together and moving the truncation error terms to the right side:

$$\begin{aligned} u_t + au_x =& \frac{\Delta t}{2}(a^2u_{xx} - u_{tt}) + \left[\frac{a\Delta x^2}{3} - \frac{a^2\Delta t\Delta x}{2}\right]u_{xxx} - \frac{\Delta t^2}{6}u_{ttt} \\ &+ \left[\frac{7a^2\Delta t\Delta x^2}{24} - \frac{a\Delta x^3}{4}\right]u_{xxxx} - \frac{\Delta t^3}{24}u_{tttt} \end{aligned} \tag{8}$$

Now, we need to replace the time derivatives ($u_{tt}, u_{ttt}$, etc.) with spatial derivatives.
Using the simple first-order relation $u_t \approx -au_x$, we can easily get:

$$u_{ttt} \approx -a^3u_{xxx} \tag{9}$$

$$u_{tttt} \approx a^4u_{xxxx} \tag{10}$$

To replace the $(a^2u_{xx} - u_{tt})$ term without losing our 3rd-order accuracy, we need to include the leading error. Let $u_t = -au_x + \mu u_{xxx}$. Taking the time derivative gives us:

$$u_{tt} = -au_{xt} + \mu u_{xxxt} = -a(-au_{xx} + \mu u_{xxxx}) - a\mu u_{xxxx} = a^2u_{xx} - 2a\mu u_{xxxx} \tag{11}$$

So, the difference is just:

$$\frac{\Delta t}{2}(a^2 u_{xx} - u_{tt}) = a\mu \Delta t u_{xxxx} \tag{12}$$

Let $C = \frac{a\Delta t}{\Delta x}$. We first collect all the $u_{xxx}$ terms (which are $\mathcal{O}(\Delta^2)$) to find $\mu$ and our coefficient $A$:

$$A \cdot a\Delta x^2 \cdot u_{xxx} = \mu \cdot u_{xxx} = \left(\frac{a\Delta x^2}{3} - \frac{a^2 \Delta t \Delta x}{2} + \frac{a^3 \Delta t^2}{6}\right) u_{xxx} \tag{13}$$

Factoring out $a\Delta x^2$:

$$A = \frac{1}{3} - \frac{1}{2}C + \frac{1}{6}C^2 \tag{14}$$

$$A = \frac{(C-1)(C-2)}{6} \tag{15}$$

Next, we collect all the $u_{xxxx}$ terms (which are $\mathcal{O}(\Delta^3)$) to find $B$:

$$B \cdot a\Delta x^3 = a\mu \Delta t + \frac{7a^2 \Delta t \Delta x^2}{24} - \frac{a\Delta x^3}{4} - \frac{a^4 \Delta t^3}{24} \tag{16}$$

Divide everything by $a\Delta x^3$ and plug in $C$:

$$B = \frac{\mu \Delta t}{\Delta x^3} + \frac{7}{24}C - \frac{1}{4} - \frac{1}{24}C^3 \tag{17}$$

Since we already know $\frac{\mu \Delta t}{\Delta x^3} = C \cdot A = \frac{C(C-1)(C-2)}{6}$, we can substitute that in:

$$\begin{aligned} B &= \frac{C(C^2 - 3C + 2)}{6} + \frac{7C - 6 - C^3}{24} \\ &= \frac{4C^3 - 12C^2 + 8C + 7C - 6 - C^3}{24} \end{aligned} \tag{18}$$

$$B = \frac{3C^3 - 12C^2 + 15C - 6}{24} = \frac{C^3 - 4C^2 + 5C - 2}{8} \tag{19}$$

Factoring the top part gives us our final expression for $B$:

$$B = \frac{(C-1)^2(C-2)}{8} \tag{20}$$

Because there are no first-order error terms (like $\mathcal{O}(\Delta x)$ or $\mathcal{O}(\Delta t)$ in the modified equation), this proves the scheme is second-order accurate in both space and time.

# Question 16: Derivation of the Modified Equation

We consider the Lax scheme for the 1D advection equation:

$$\frac{u_j^{n+1} - \frac{u_{j+1}^n + u_{j-1}^n}{2}}{\Delta t} + \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0$$

Define the CFL number:

$$C = \frac{a\Delta t}{\Delta x}$$

## (a) Von Neumann Stability Analysis

Assume a Fourier mode:

$$u_j^n = G^n e^{ikj\Delta x}$$

Substitute into the scheme:

$$G = \frac{1}{2}\left(e^{ik\Delta x} + e^{-ik\Delta x}\right) + \frac{a\Delta t}{2\Delta x}\left(e^{ik\Delta x} - e^{-ik\Delta x}\right)$$

Using Euler identities:

$$G = \cos(k\Delta x) - iC\sin(k\Delta x)$$

Compute the amplification factor:

$$|G|^2 = \cos^2(k\Delta x) + C^2\sin^2(k\Delta x)$$

For stability, require:

$$|G| \leq 1 \quad \forall k$$

Thus:

$$C^2 \leq 1$$

$$\boxed{|C| \leq 1}$$

## (b) Taylor Expansion and Order of Accuracy

Time term:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = u_t + \frac{\Delta t}{2} u_{tt} + O(\Delta t^2)$$

Averaging term:

$$\frac{u_{j+1} + u_{j-1}}{2} = u + \frac{\Delta x^2}{2} u_{xx} + O(\Delta x^4)$$

Central difference:

$$\frac{u_{j+1} - u_{j-1}}{2\Delta x} = u_x + \frac{\Delta x^2}{6} u_{xxx} + O(\Delta x^4)$$

Substitute into the scheme:

$$u_t + \frac{\Delta t}{2} u_{tt} + a\left(u_x + \frac{\Delta x^2}{6} u_{xxx}\right) = \frac{\Delta x^2}{2\Delta t} u_{xx}$$

Rearrange:

$$u_t + a u_x = \frac{\Delta x^2}{2\Delta t} u_{xx} + \frac{\Delta t}{2} u_{tt} + \frac{a\Delta x^2}{6} u_{xxx}$$

Using the PDE:

$$u_t = -a u_x$$

we obtain:

$$u_{tt} = a^2 u_{xx}$$

Thus:

$$u_t + a u_x = \left(\frac{\Delta x^2}{2\Delta t} - \frac{a^2 \Delta t}{2}\right) u_{xx} + \frac{a\Delta x^2}{6} u_{xxx}$$

**Order of Accuracy**:

- Time: first-order
- Space: second-order

## (c) Modified Equation (First-order Error Only)

Retain only the leading error term:

$$u_t + au_x = \alpha\Delta x, u_{xx}$$

From part (b):

$$\alpha\Delta x = \frac{\Delta x^2}{2\Delta t} - \frac{a^2\Delta t}{2}$$

Divide by $\Delta x$:

$$\alpha = \frac{\Delta x}{2\Delta t} - \frac{a^2\Delta t}{2\Delta x}$$

Using the CFL number:

$$C = \frac{a\Delta t}{\Delta x}$$

we obtain:

$$\alpha = \frac{a}{2}\left(\frac{1}{C} - C\right)$$

Stability Condition from Modified Equation

For numerical stability, require:

$$\alpha \geq 0$$

$$\frac{1}{C} - C \geq 0 \Rightarrow C^2 \leq 1$$

$$\boxed{|C| \leq 1}$$